OBJECT ORIENTED PROGRAMMING

ACTIVITY #3 – Basic ATM System

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BSIT 2-3

Objective:

Create a Java program that accepts student information and evaluates the student's grade using decision structures.

Program Requirements

1. Create a Java program named BasicATMSystemLastName.
2. Use the Scanner class to accept user input.
3. Assume the initial balance is 5,000 pesos.

Display the Menu

===== ATM MENU =====

1. Check Balance

2. Deposit

3. Withdraw

4. Exit

Program Rules

- Use a switch statement to process the selected option.
- Use if–else statements to validate withdrawal.
- Maximum withdrawal per transaction: 2,000 pesos.
- The program must repeat using a do-while loop until the user selects Exit.

Submission Format

- Submit the Java source code file (.java).
- The filename must follow this format:
  Section_LastName_FirstName_ActivityName.java
- Include screenshots of the program output showing the following:
  - User input
  - Program processing
  - Final output/result
- The screenshot must clearly display the entire console output.

Source Code:

```java
import java.util.Scanner;

public class BSIT23_Rivas_DaveAillerr_BasicATMSystem {
    Run | Debug
    public static void main(String[] args) {
        // Declare input scanner
        Scanner input = new Scanner(System.in);

        // Variable declaration
        int answer = 0;
        double withdraw = 0;
        double deposit = 0;
        double currentBalance = 5000;

        do {
            // Prints Menu
            System.out.println(x: "==== ATM MENU ====");
            System.out.println(x: "1. Check Balance");
            System.out.println(x: "2. Deposit");
            System.out.println(x: "3. Withdraw");
            System.out.println(x: "4. Exit");

            // Ask user for selecting options
            System.out.print(s: "\nPlease enter your choice: ");
            answer = input.nextInt();

            switch (answer) {
                case 1:
                    // Check current balance
                    System.out.println(x: "==== Checking your current balance ====");
                    System.out.println("Current Balance: " + currentBalance);
                    System.out.println(x: "=====================================\n");
                    break;
                case 2:
                    // Deposit logic
                    System.out.println(x: "==== Depositing Money ====");
                    System.out.print(s: "Enter deposit amount: ");
```

```java
                    // Deposit logic
                    System.out.println(x: "==== Depositing Money ====");
                    System.out.print(s: "Enter deposit amount: ");
                    deposit = input.nextDouble();
                    currentBalance += deposit; // Add deposit to current balance
                    System.out.println(x: "=====================================");
                    System.out.println("Deposit Successful! New Balance: " + currentBalance);
                    System.out.println(x: "=====================================\n");
                    break;
                case 3:
                    // Withdrawal logic
                    System.out.println(x: "==== Withdrawing Money ====");
                    System.out.print(s: "Enter withdraw amount: ");
                    withdraw = input.nextDouble();

                    if (withdraw > 2000) {
                        System.out.println(x: "=====================================");
                        System.out.println(x: "Maximum withdrawal exceeded. (2000 pesos)");
                        System.out.println(x: "=====================================\n");
                    } else if (currentBalance < withdraw) {
                        System.out.println(x: "=====================================");
                        System.out.println(x: "Not enough balance to withdraw.");
                        System.out.println(x: "=====================================\n");
                    } else {
                        currentBalance = currentBalance - withdraw;
                        System.out.println(x: "=====================================");
                        System.out.println("Withdrawal Successful! New Balance: " + currentBalance);
                        System.out.println(x: "=====================================\n");
                    }
                    break;
                case 4:
                    System.out.println(x: "=====================================");
                    System.out.println(x: "                Exiting... ");
                    System.out.println(x: "=====================================\n");
                    break;
```

```java
                System.out.println(x: "                Exiting... ");
                System.out.println(x: "=====================================\n");
                break;
            default:
                System.out.println(x: "Please enter a valid choice.");
                break;
        }
    } while (answer != 4);
    input.close();
  }
}

```

Input & Output:

```
==== ATM MENU ====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Please enter your choice:
```

```
Please enter your choice: 1
==== Checking your current balance ====
Current Balance: 5000.0
========================================
```

```
Please enter your choice: 2
==== Depositing Money ====
Enter deposit amount: 5000
========================================
Deposit Successful! New Balance: 10000.0
========================================
```

```
Please enter your choice: 3
==== Withdrawing Money ====
Enter withdraw amount: 5000
========================================
Maximum withdrawal exceeded. (2000 pesos)
========================================
```

```
Please enter your choice: 3
==== Withdrawing Money ====
Enter withdraw amount: 2000
========================================
Withdrawal Successful! New Balance: 8000.0
========================================
```

```
Please enter your choice: 1
==== Checking your current balance ====
Current Balance: 8000.0
========================================
```

```
==== ATM MENU ====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Please enter your choice: 4
========================================
                Exiting...
========================================
```